



Hantro SW VC9000E

Release Notes

Sep. 2024

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1 General Information

This guide provides an overview of the VC9000E SW package, including some example binary running command lines and sample compile commands, for customer to kick off the sw package quickly.

If need more details, please refer to the documents listed in Chapter 2 "Documents List".

Version: 2.4.123

Release Date: Sep.27.2024

Release Package Type: CtrlSW_Src

Release Formats: h264/hevc/jpeg

CModel Built Environment:

ldd version: ldd (Ubuntu GLIBC 2.23-0ubuntu10) 2.23

Linux version: 4.4.0-62-generic (buildd@lcy01-30)

gcc version: 5.4.0 20160609 (Ubuntu 5.4.0-6ubuntu1~16.04.4)

2 Package Contents List

```
|—— doc
|   |—— Hantro.VC9000E.JPEG.Encoder.API.User.Manual-v2.4.123-A-20240919.pdf
|   |—— Hantro.VC9000E.JPEG.Encoder.TestBench.User.Manual-v2.4.123-A-20240919.pdf
|   |—— Hantro.VC9000E.Series.Video.Encoder.FPGA.Verification.Guide-v1.10-D-20221102.pdf
|   |—— Hantro.VC9000E.Video.Encoder.API.User.Manual-v2.4.123-A-20240919.pdf
|   |—— Hantro.VC9000E.Video.Encoder.TestBench.User.Manual-v2.4.123-A-20240919.pdf
|   |—— Hantro.VC9000E.Video.Encoder Wrapper.Layer.API.User.Manual-v2.4.123-A-
20240919.pdf
|   |—— license
|   |—— Hantro.Programming.Memo.VPU.Encoder.Software.Source.License.Summary-v0.80-D-
20240515.pdf
|—— Hantro_SW_Encoder_Cmodel_Bin_Proprietary_v.2.4.123_CL981740P_20240923.tgz
|—— Hantro_SW_Encoder_CtrlSW_Proprietary_v.2.4.123_CL981740P_20240923.tgz
|—— Hantro_SW_Encoder_example_test_command_20240927.tgz
|—— Hantro_SW_Encoder_Kernel_Redistributable_v.2.4.123_CL981740_20240923.tgz
|—— Hantro_SW_VC9000E_Test_Report_basedon_CL981740P_20240927.tgz
```

2 directories, 12 files

3 Package Structure

3.1 The main structure of the VC9000E_CtrlSW package

Hantro_SW_Encoder_Cmodel_Bin_Proprietary_v.2.4.123_CL981740P_20240923.tgz. It's the main structure for vc9000e cmodel binaries.

```
VC9000E_CtrlSW_20240923/
├── binary
│   ├── ASM_n
│   │   ├── h264_testenc_vce
│   │   ├── hevc_testenc_vce
│   │   └── jpeg_testenc_vce
│   └── ASM_y
│       ├── h264_testenc_vce
│       ├── hevc_testenc_vce
│       └── jpeg_testenc_vce
```

binary Cmodel binaries, ASM=y means compile binary with ASM=y, which can improve testing speed, default ASM=n. The old version of CPU may not support ASM_y binaries, customer can use ASM_n bin if meeting some compatibility issues with ASM_y tools.

3.2 The main structure of the VC9000E_CtrlSW package

Hantro_SW_Encoder_CtrlSW_Proprietary_v.2.4.123_CL981740P_20240923.tgz. It's the main structure for vc9000e ctrlsw, including user space code.

```
VC9000E_CtrlSW_20240923
├── software
│   ├── build
│   ├── inc
│   ├── linux_reference
│   ├── Makefile
│   └── source
├── system
└── models
```

software	including sw user space code
--build	some build rules for Makefile
--inc	some header files
--linux_reference	a folder reference drivers and testbenches
--Makefile	binary compilation files
--source	source code to generate "libh2enc.a"

system some header files

3.3 The main structure of the Kernel package

Hantro_SW_Encoder_Kernel_Redistributable_v.2.4.123_CL981740_20240923.tgz. It's about the kernel driver source code

VC9000E_CtrlSW_20240923/software/linux_reference/

├── kernel_module

└── memalloc

kernel_module linux kernel driver code

memalloc memory allocation kernel driver

3.4 The main structure of the example test package

Hantro_SW_Encoder_example_test_command_20240927.tgz. It's about the running command for customer reference.

VC9000E_example_test_command

└── new_vce_command_custom.sh

new_vce_command_custom.sh VC9000E Regular Example test commands

3.5 The main structure of the test results

Hantro_SW_Encoder_example_test_command_20240927.tgz. It's about the test results for customer reference.

4 Package Assemble(MUST)

Please untar the 3.2 VC9000E_CtrlSW package and 3.3 kernel package, and then the structure of VC9000E_CtrlSW_20240905/software/linux_reference path will like below. **This step is necessary before doing below compile steps!!!**

linux_reference

├── Baseaddress

├── debug_trace

├── ewl

├── kernel_module

├── Makefile

├── memalloc

└── test

5 How to load pcie driver

```
$> cd VC9000E_CtrlSW_20240923/software/linux_reference/kernel_module/linux
$> make clean && make SUPPORT_MMU=y SUPPORT_48PA_MMU=y
$> ./driver_load.sh vcmd=1
$> cd VC9000E_CtrlSW_20240923/software/linux_reference/memalloc
$> make clean && make
$> ./memalloc_load.sh
```

6 How to compile VC9000E pcie test binary in FPGA ENV

Use h264 as reference, and it's similar to other formats.

Normal:

```
$> cd VC9000E_CtrlSW_20240923/software
$> make clean h264 ENV=pci SUPPORT_MMU=y SUPPORT_48PA_MMU=y
```

Dec400:

```
$> cd VC9000E_CtrlSW_20240923/software
$> make clean h264 ENV=pci SUPPORT_MMU=y SUPPORT_48PA_MMU=y SUPPORT_DEC400=y
```

Then you will find pcie test binary in ./bin/pci.

7 How to run the binaries

Here are some sample command lines for reference

7.1 For CModel

```
$> ./h264_testenc_vce -w 352 -h 288 -b 19 --gopSize=1 -i foreman_cif.yuv -o stream.h264
#-w 352 -h 288: source image is 352x288
#-b 19: the last picture of the input file is frame 19, so 20 frames will be encoded
#-i foreman_cif.yuv: Set the input file
#-o stream.h264: Set the output stream file
```

7.2 For FPGA

Please refer to CModel command lines.

More details about parameters for different functions please run "hevc_testenc_vce", or refer to doc/Hantro.VC9000E.V1x.* Encoder.TestBench*.pdf.